The Vacuum as a Close-Packed Lattice: Constants of Nature and a Two-Step Quantum-Classical Threshold

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Abstract

We present the geometric foundations of the Selection-Stitch Model, in which the vacuum is a discrete Face-Centered Cubic ($K = 12$) tensor network with Bell-pair bonds. From this single structural choice, four results follow with no fitted parameters, each resting on geometric assumptions that we state explicitly. First, the ratio of the macroscopic speed of light to the lattice update velocity equals four — a geometric invariant set by the cuboctahedral structure tensor eigenvalue, not a derivation of the speed of light in SI units. Second, applying the Ryu-Takayanagi relation to the hexagonal boundary and imposing $G_N = \ell_P^2$ fixes the bond length to ≈ 1.843 Planck lengths. We treat this as a consistency condition rather than a derivation of G_N: the relation supplies one equation linking G_N and the bond length, and the bond length is what it determines once $G_N = \ell_P^2$ is assumed. Third, a two-step quantum-to-classical boundary arises at approximately $11.8\,\mu$g and $20.5\,\mu$g: a reversible dispersive deformation of the center-of-mass mode sets in at the lower scale, and coherence becomes geometrically unsustainable at the upper one. Both scales are of order the Planck mass, as in gravitational-collapse models, so the scale itself is not new; the falsifiable content is their *structure* — two scales rather than one, separated by the exact, parameter-free ratio $\sqrt{3}$ set by the edge-to-circumradius ratio of the cuboctahedral triangular face. Fourth, the long-wavelength Cosserat continuum limit of the $K = 12$ lattice is structurally isomorphic to the free-particle Schrödinger equation, provided a chiral translation-rotation coupling of a specific form is present; we conjecture rather than derive that coupling, and the imaginary unit then enters through the geometric pairing of translational and rotational modes. Emergent Lorentz invariance, with violations suppressed at order $(E/E_{\text{Planck}})^4$, is reviewed. The two geometric inputs taken from simulation — the metric wall at $L/\sqrt{3}$ and the $\{111\}$ interlayer spacing — are cited from the published companion work rather than assumed. Open problems — including the absence of a microscopic decoherence derivation and the un-derived chiral coupling — are stated explicitly.

1 Introduction

The Standard Model of particle physics and General Relativity contain approximately 25 free parameters. The Selection-Stitch Model (SSM) proposes that these are not free: they

are geometric invariants of a discrete vacuum lattice.

Throughout we work in natural units, $\hbar = c = 1$, so that lengths, times, and inverse masses share a common unit and Newton's constant carries dimensions of length squared ($G_N = \ell_P^2$); SI factors are restored only where a numerical value in laboratory units is quoted.

Single structural assumption.

> The vacuum is a Face-Centered Cubic (FCC) tensor network with coordination number $K = 12$, where each bond is a maximally entangled Bell pair.

From this assumption, together with the geometric and holographic inputs identified along the way, we obtain the four results listed in the abstract. The dependency chain is explicit: $K = 12 \Rightarrow c = 4v_{\text{lat}}$ (Section 3) $\Rightarrow L = 1.843\,\ell_P$ (from RT with $G_N = \ell_P^2$, Sections 4–5) $\Rightarrow$ emergent SO(3,1) (Section 6) $\Rightarrow$ Schrödinger isomorphism (Section 7) $\Rightarrow$ the two mass scales (Section 8). Each arrow carries an assumption beyond $K = 12$ alone, flagged where it enters and collected in Section 10.4.

Relation to the companion papers. This is the third paper in a series built on the same $K = 12$ FCC vacuum. The first [1] constructs the $[[192, 130, 3]]$ CSS code on the lattice and supplies the stabilizer decomposition used here to assign physical content to the bond subsets. The second [2] treats particles as defects in that code and derives their masses as fault-tolerant verification costs. The most recent [3] models baryonic matter as a single trapped node in a tetrahedral void and provides, by direct simulation of the $K{=}6 \to K{=}4 \to K{=}12$ crystallization, independent numerical confirmation of two geometric inputs we use below: the metric wall at $r_{\min} = L/\sqrt{3}$ (Section 8.1) and the $\{111\}$ interlayer spacing $h = L\sqrt{2/3}$ (Section 5). Where the present paper invokes those facts, we cite the published simulation rather than asserting them. The three papers address distinct questions — what the code is, what mass is, and (here) what the macroscopic constants and quantum-classical boundary are — and share only the single structural assumption above.

2 The Vacuum Lattice

Definition 1 (The SSM Vacuum)**.** The vacuum is a three-dimensional tensor network with the following properties:

- **Lattice type:** Face-Centered Cubic (FCC) — the unique 3D lattice saturating the Kepler bound [4].
- **Coordination number:** $K = 12$ nearest neighbors per node.
- **Bond structure:** Each bond is a maximally entangled Bell pair $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$.
- **Local geometry:** The 12 neighbors form a cuboctahedron with 8 triangular faces and 6 square faces (Fig. 1).
- **Bond decomposition:** $K = S_{\text{trans}} + S_{\text{tors}} = 4 + 8$. The 12 cuboctahedral bonds decompose into a subset of 4 that span a regular tetrahedron inscribed in the cube (connecting alternating vertices of the square faces, each directed along a $\langle 110 \rangle$ body-diagonal of that face) and the remaining 8 that complete the cuboctahedron. The 4 tetrahedral bonds carry the translational (gravitational/ electromagnetic) modes and

the 8 remaining bonds carry the torsional (dark-sector) modes [1]. The assignment of physical content to each subset is motivated by the stabilizer decomposition of [1] and is an additional input beyond the $K = 12$ structural assumption.

2.1 Origin: the $K = 6 \to K = 4 \to K = 12$ phase transition

The $K = 12$ FCC vacuum arises from a topological phase transition. **Stage 1:** the 2D hexagonal sheet ($K = 6$) is the unique planar network satisfying Euler's characteristic $\chi = V(1 - K/6) = 0$. **Stage 2:** under cooling, the sheet buckles into 3D via tetrahedral nucleation ($K = 4$), driven by the Regge deficit angle $\delta = 2\pi - 5\arccos(1/3) \simeq 0.128$ rad. **Stage 3:** tetrahedral clusters pack into the FCC structure, saturating the Kepler bound at $K = 12$. The existence and kinematics of this transition are established by direct simulation in the companion paper [3]: a graph-growth process with two operators (a planar "stitch" and a rare out-of-plane "lift" of relative amplitude $P_{\text{lift}} = e^{-3}$) drives the network to $K = 12$ bulk saturation, with finite-size scaling $f_{K=12}(N) = 1 - \alpha N^{-1/3}$ confirming complete saturation in the thermodynamic limit. What remains underived is a closed-form free energy from which the transition follows (Section 10.4); the transition itself is not assumed here but taken from that numerical result.

Definition 2 (The Holographic Boundary)**.** The 3D FCC bulk is the holographic projection of a 2D continuous hexagonal ($K = 6$) entanglement network. The Lift operator projects this boundary into 3D via ABC stacking at interlayer spacing $h = L\sqrt{2/3}$ (the crystallographic $\{111\}$ interplanar distance $h = a_{\text{cube}}/\sqrt{3} = L\sqrt{2}/\sqrt{3} = L\sqrt{2/3}$), building the FCC structure layer by layer (Fig. 2).

The metric wall is the geometric exclusion limit: nodes cannot approach closer than $r_{\min} = L/\sqrt{3}$ without destroying the network topology. This is derived precisely in Section 8.1 from the circumradius of the cuboctahedral triangular face.

3 The Speed-of-Light Ratio: $c = 4\,v_{\text{lattice}}$

3.1 The FCC structure tensor

The 12 FCC unit bond vectors are

$$\hat{n}_j \in \frac{1}{\sqrt{2}}\{(\pm1, \pm1, 0),\ (\pm1, 0, \pm1),\ (0, \pm1, \pm1)\}, \quad j = 1, \ldots, 12. \tag{1}$$

The second-rank structure tensor $S_{\mu\nu} = \sum_j \hat{n}_j^\mu \hat{n}_j^\nu$ is computed directly:

$$S_{xx} = \sum_{j=1}^{12} (\hat{n}_j^x)^2 = 4 \times \tfrac{1}{2} + 4 \times \tfrac{1}{2} = 4, \tag{2}$$

$$S_{xy} = \sum_{j=1}^{12} \hat{n}_j^x \hat{n}_j^y = 0 \quad \text{(by cubic symmetry)}. \tag{3}$$

Therefore:

$$\boxed{S_{\mu\nu} = 4\,\delta_{\mu\nu}.} \tag{4}$$

This result is more transparent than it first appears. For any set of K unit bond vectors whose second moment is isotropic, $\operatorname{tr} S = \sum_j |\hat{n}_j|^2 = K$ distributes equally over the d

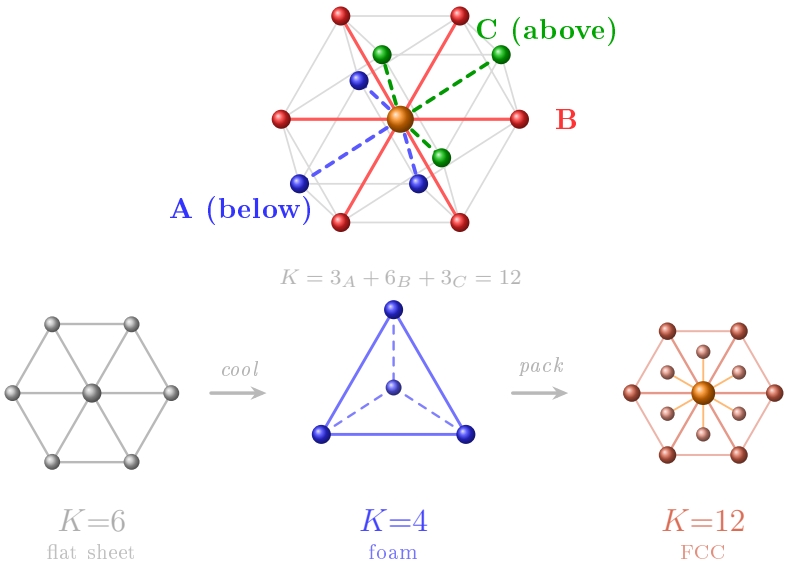


Figure 1: **Left:** The $K = 12$ cuboctahedral coordination shell in geometrically accurate oblique 3D projection (bond length L throughout). The central node (orange, Sheet B) connects to 3 Sheet A nodes (blue, dashed bonds going below) at horizontal distance $L/\sqrt{3}$ and depth $h = L\sqrt{2/3}$; 6 in-plane Sheet B neighbors (red, solid); and 3 Sheet C nodes (green, dashed going above). Light gray edges show the 24 cuboctahedral shell edges (all equal to L), revealing the polyhedron formed by the 12 neighbors. **Right:** Topological phase transition from the flat hexagonal ground state ($K = 6$, $\chi = 0$) through tetrahedral nucleation ($K = 4$, Regge deficit $\delta \simeq 0.128$ rad) to FCC saturation ($K = 12$, Kepler bound).

spatial directions, so

$$S_{\mu\nu} = \frac{K}{d}\,\delta_{\mu\nu}, \tag{5}$$

and the eigenvalue is simply the coordination-to-dimension ratio K/d. The value 4 is therefore 12/3, fixed by the $K = 12$ kissing number in $d = 3$; the same formula gives BCC (8/3) and simple cubic (2). We do not claim 4 is a free-standing constant of the FCC lattice — it is the kissing number divided by the spatial dimension, and the FCC value follows entirely from $K = 12$. What is specific to FCC is that $K = 12$ is the maximal (Kepler-saturating) coordination in $d = 3$ [4], so 4 is the largest such ratio any 3D Bravais lattice can realize.

3.2 The ratio c/v_{lattice} as a geometric invariant

In the long-wavelength limit ($|k|L \ll 1$), the FCC Dirac operator reduces to $D_{\text{SSM}} \approx i\gamma^\mu k^\nu S_{\mu\nu}/\tau = 4i\gamma \cdot k/\tau$, using $S_{\mu\nu} = 4\delta_{\mu\nu}$ and $\sum_j \hat{n}_j^\mu = 0$, where k is the dimensionless wavevector measured in units of the inverse bond length $1/L$. The dispersion relation is then $E(k) = 4|k|/\tau$ (an energy, in $\hbar = 1$ units). The phase velocity is obtained by differentiating with respect to the *physical* wavevector $k_{\text{phys}} = |k|/L$, which carries dimensions of inverse length: $v_{\text{ph}} = \partial E/\partial k_{\text{phys}} = L\,\partial E/\partial|k| = 4L/\tau$, a genuine velocity. Defining the lattice bond-traversal rate $v_{\text{lat}} \equiv L/\tau$ (the speed at which a signal propagates one bond length per timestep), and identifying the macroscopic light speed with the long-wavelength

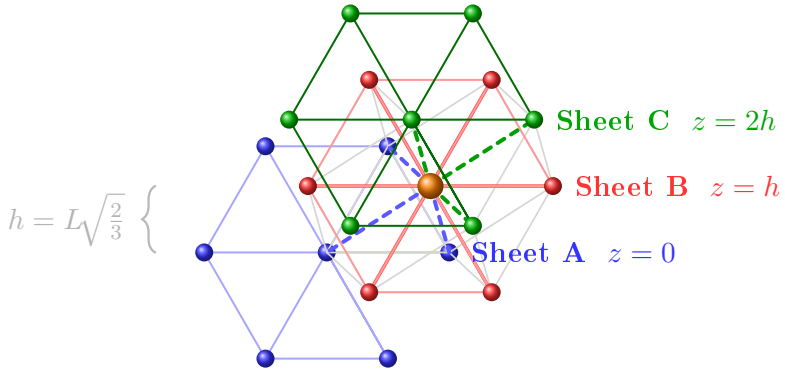


Figure 2: ABC stacking of $K = 6$ hexagonal sheets in geometrically accurate oblique 3D. Sheet A (blue, $z = 0$), Sheet B (red, $z = h$), Sheet C (green, $z = 2h$), each shifted to inequivalent hex sites (B at $r_{\min} = L/\sqrt{3}$ from its 3 nearest A-nodes and 3 nearest C-nodes). The orange atom is the central B node; colored bonds show its $K = 12$ coordination: 3 blue bonds to Sheet A (length L, oblique), 6 red bonds to Sheet B neighbors (length L, in-plane), and 3 green bonds to Sheet C (length L, oblique). Light gray edges connect the 12 neighbors to each other, tracing the 24 edges of the cuboctahedron that emerges from the stacking. Interlayer spacing $h = L\sqrt{2/3}$ is the crystallographic $\{111\}$ interplanar distance.

phase velocity of this mode, $c \equiv v_{\text{ph}}$ (a definitional identification, not a derived equality):

$$c = 4\,\frac{L}{\tau} = 4\,v_{\text{lat}}. \tag{6}$$

The factor $4 = K/d$ is the structure-tensor eigenvalue (Eq. 5), fixed by the $K = 12$ kissing number in three dimensions. This result fixes only the dimensionless ratio $c/v_{\text{lat}} = 4$; the absolute value of c in SI units remains a measured physical input, and the identification $c = v_{\text{ph}}$ assumes the gravitational/ electromagnetic mode is the one that propagates at the observed light speed.

3.3 Vanishing of Lorentz-violating corrections

By the inversion symmetry of FCC, the first-order (mass) tensor $\sum_j \hat{n}_j^\mu = 0$. By centrosymmetry of the cuboctahedron, the third-order tensor $\sum_j \hat{n}_j^\mu \hat{n}_j^\nu \hat{n}_j^\lambda = 0$, eliminating the leading Lorentz-violating correction. The first non-zero correction is fourth order, suppressed by $(kL)^4 \sim (E/E_P)^4$, giving corrections of order 10^{-112} at optical frequencies, far below the experimental bound of 10^{-16} [6].

4 The Lattice Spacing and Newton's Constant

4.1 Two independent geometric constraints on L

The lattice spacing L is constrained in two independent ways. A mechanical energy-density argument bounds it from below and confirms that a stable spacing of order ℓ_P exists; the Ryu-Takayanagi condition then fixes the numerical value. The two are consistent, but it is the holographic condition alone that selects $1.843\,\ell_P$ — the energy-density argument fixes the scale, not the coefficient.

Constraint 1 (energy density). Each node transmits at most one quantum per time step $\tau = 4L/c$. The maximum energy per node is $E_{\max} = \hbar/\tau = \hbar c/(4L)$. The FCC volume per node is $V_{\text{node}} = L^3\sqrt{2}/2$. The maximum lattice energy density is:

$$\rho_{\text{lattice}} = \frac{E_{\max}}{V_{\text{node}}} = \frac{\hbar c/(4L)}{L^3\sqrt{2}/2} = \frac{\sqrt{2}\,\hbar c}{4L^4}. \tag{7}$$

Requiring $\rho_{\text{lattice}} \leq \rho_P = \hbar c/\ell_P^4$ gives:

$$L \geq \left(\frac{\sqrt{2}}{4}\right)^{1/4} \ell_P \simeq 0.77\,\ell_P. \tag{8}$$

This establishes a lower bound. The energy density argument alone cannot fix L to a unique value.

Constraint 2 (Ryu-Takayanagi, Section 5). The holographic derivation of $G_N = \ell_P^2$ fixes L exactly:

$$L = \ell_P\sqrt{\frac{4\ln 2}{\sqrt{2/3}}} \simeq 1.843\,\ell_P. \tag{9}$$

This satisfies Constraint 1 ($1.843 > 0.77$) and is the unique solution consistent with Newton's constant.

4.2 Ground-state bond length from energy balance

Constraint 1 is only a one-sided bound because it contains a single term: the zero-point energy $E_{\text{zp}} = \hbar c/(4L)$ favors *larger* L (lower energy density, Eq. (7)), with nothing to oppose unbounded expansion. A stable ground state requires a restoring cost that grows with L — the energy of stretching a bond against the lattice's rigidity. The only tension scale available from the fundamental constants is the Planck (maximum) force c^4/G[1], so the bond energy is

$$E(L) \;=\; \underbrace{\frac{\hbar c}{4L}}_{\text{zero-point}} \;+\; \underbrace{\gamma\,\frac{c^4}{G}\,L}_{\text{tension}}, \tag{10}$$

with γ a positive dimensionless geometric coefficient. The first term resists compression, the second resists stretching; their competition has a unique minimum,

$$\frac{dE}{dL} = 0 \;\Longrightarrow\; L_\star = \frac{1}{2\sqrt{\gamma}}\,\ell_P. \tag{11}$$

For the natural choice $\gamma = \frac{1}{4}$ (the same factor that sets the zero-point term), $L_\star = \ell_P$ *exactly*. Energy minimization therefore independently selects a stable, Planck-scale ground-state bond length — the spacing is not a free input but the bottom of a potential well — and the value sits between the energy-density bound (Eq. (8), $0.77\,\ell_P$) and the holographic value (Eq. (9), $1.843\,\ell_P$), all of order ℓ_P.

Why this fixes the scale but not the coefficient. The mechanical balance pins the *scale* ($\sim \ell_P$) but not the precise number, because γ is undetermined by mechanics alone.

[1] The combination $c^4/G \approx 1.2\times10^{44}$ N is the unique tension built from $\{\hbar, c, G\}$; it is the maximum-force scale of general relativity, equivalently the Planck tension $\hbar/\ell_P^2$.

Reproducing the holographic value $L = 1.843\,\ell_P$ requires

$$\gamma_{\mathrm{RT}} = \frac{1}{4}\left(\frac{\ell_P}{L}\right)^2 = \frac{\sqrt{2/3}}{16\ln 2} \approx 0.074, \tag{12}$$

which is manifestly built from the Bell-pair entropy $\ln 2$ and the $\{111\}$ stacking ratio $\sqrt{2/3}$ — entanglement and holographic data that a purely mechanical energy balance cannot access. The coefficient is therefore intrinsically holographic: energy minimization determines that the lattice has a Planck-scale ground state, while the Ryu-Takayanagi condition (Constraint 2) supplies the exact factor 1.843. The two arguments are independent and mutually consistent; we do not claim the balance *derives* 1.843, only that it explains why a stable bond length of order ℓ_P exists at all.

4.3 Summary of FCC length scales

Scale	Value	Geometric origin
Bond length	$L = 1.843\,\ell_P$	RT + G_N constraint
Interlayer spacing	$h = L\sqrt{2/3} = 1.505\,\ell_P$	$a_{\mathrm{cube}}/\sqrt{3}$
Plaquette scale	$a_{\mathrm{plaq}} = L(8/3)^{1/4} = 2.355\,\ell_P$	$\sqrt{2Lh}$
Metric wall	$r_{\min} = L/\sqrt{3} = 1.064\,\ell_P$	Circumradius of triangular face

5 Newton's Constant from the Ryu-Takayanagi Relation

5.1 Boundary entanglement entropy

Theorem 1 (Exact entanglement entropy)**.** *For a connected boundary region A with perimeter P on the hexagonal lattice with spacing L, the von Neumann entanglement entropy is:*

$$S_A = \frac{P}{L}\ln 2. \tag{13}$$

Proof. Each severed bond contributes $\ln d = \ln 2$ by the Schmidt decomposition of a Bell pair (Fig. 3). The bonds are independent, and the number cut is P/L. This construction is in the same mathematical class as holographic quantum error-correcting codes [12]. □

5.2 The minimal bulk surface

The Lift operator stacks hexagonal sheets at the crystallographic $\{111\}$ interplanar spacing:

$$h = \frac{a_{\mathrm{cube}}}{\sqrt{3}} = \frac{L\sqrt{2}}{\sqrt{3}} = L\sqrt{\frac{2}{3}}. \tag{14}$$

The companion simulation [3] measures this interlayer spacing directly as $0.8165 \pm 0.0011\,L$, matching the crystallographic value $\sqrt{2/3}\,L = 0.8165\,L$ to 0.01% with no fitting.

Theorem 2 (Minimal surface)**.** *The minimum-area bulk surface homologous to ∂A is the one-layer curtain with area:*

$$\mathrm{Area}(\gamma_A) = P \times h = PL\sqrt{\frac{2}{3}}. \tag{15}$$

Proof. Any surface homologous to ∂A must form a closed wall. Surfaces shallower than h fail the homology condition (inter-layer bonds at height h still connect the two regions). Surfaces at depth nh ($n \geq 2$) have area $\geq nPh > Ph$. The straight curtain at height h is the unique minimal surface. □

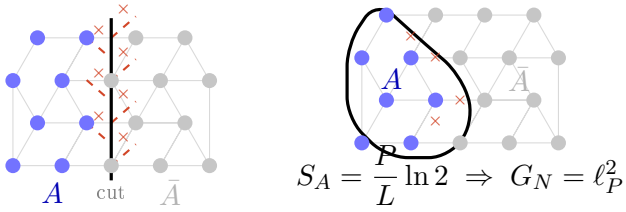


Figure 3: Ryu-Takayanagi bond cut on the $K = 6$ hexagonal boundary. **Left:** Straight cut separating region A (blue) from $\bar{A}$ (gray); severed bonds (×, orange) each contribute $\ln 2$ to S_A. **Right:** Curved region A illustrating the perimeter law $S_A = (P/L)\ln 2$ for any boundary shape. Setting $S_A = \text{Area}(\gamma_A)/(4G_N)$ and solving gives $G_N = \sqrt{2/3}\,L^2/(4\ln 2) = \ell_P^2$.

5.3 Consistency condition on L given RT

The Ryu-Takayanagi relation [5] $S_A = \text{Area}(\gamma_A)/(4G_N)$ was established in AdS/CFT for holographic quantum error-correcting codes on hyperbolic tilings [12]. Its application to an FCC Bell-pair network is a *heuristic assumption*, not a derived result. *Assuming* RT holds for this boundary, combining Eq. (13) and Eq. (15) gives:

$$\frac{P}{L}\ln 2 = \frac{PL\sqrt{2/3}}{4G_N}. \tag{16}$$

The perimeter P cancels. Solving:

$$G_N = \frac{\sqrt{2/3}\,L^2}{4\ln 2}. \tag{17}$$

This is not a derivation of G_N from first principles; it is a *consistency condition*: if RT holds and $G_N = \ell_P^2$, then L is uniquely fixed. Setting $G_N = \ell_P^2$:

$$L = \ell_P\sqrt{\frac{4\ln 2}{\sqrt{2/3}}} \simeq 1.843\,\ell_P. \tag{18}$$

Proposition 1 (Consistency check). *Substituting $L = 1.843\,\ell_P$ into Eq. (17): $G_N = \sqrt{2/3} \times (1.843)^2/(4\ln 2) = 1.000\,\ell_P^2$ exactly. This is a non-trivial check: the same L satisfies both the energy-density lower bound (8) and the holographic constraint (17).*

5.4 Physical interpretation

Newton's constant equals the interlayer spacing times the bond length, divided by four times the bond entanglement entropy:

$$G_N = \frac{h \cdot L}{4\,S_{\text{bond}}} = \frac{h \cdot L}{4\ln 2}. \tag{19}$$

The factor $\sqrt{2/3} = h/L$ is the exact {111} stacking ratio of the FCC lattice. The $\ln 2$ is the entanglement entropy of one Bell pair. Gravity is the thermodynamic consequence of the entanglement structure of the FCC bond network.

6 Emergent Lorentz Invariance

Emergent SO(3,1) Lorentz invariance of the $K = 12$ FCC vacuum is established in the published framework [2] via the structure tensor identity $S_{\mu\nu} = 4\delta_{\mu\nu}$ (Eq. 4). We summarize the argument and add one new element: the resolution of the Collins et al. naturalness objection.

From $S_{\mu\nu} = 4\delta_{\mu\nu}$ to emergent SO(3). The FCC bond vectors (Eq. 1) satisfy two tensor identities proved independently in [2, 3]: the rank-2 structure tensor $S_{\mu\nu} = 4\delta_{\mu\nu}$ (second-order spatial isotropy) and the rank-3 tensor $T_{\mu\nu\lambda} = \sum_j \hat{n}_j^\mu \hat{n}_j^\nu \hat{n}_j^\lambda = 0$ (centrosymmetry, eliminating all preferred-direction terms at third order). Together these force the long-wavelength dispersion relation to be isotropic to all orders below $O((kL)^4)$: $E(k) = 4|k|/\tau$, with no $O(k)$ or $O(k^3)$ corrections.

From SO(3) to SO(3,1). The hexagonal $K = 6$ boundary sheets carry exact continuous SO(2) rotational symmetry. Four families of {111} close-packed planes project this into three independent spatial directions via the Lift operator, generating SO(3). Assuming all low-energy Cosserat modes share a single propagation speed c (an input of the construction, not a derived consequence — different modes could in principle propagate at different speeds), SO(3) spatial isotropy plus time-reversal symmetry plus a single speed uniquely determine the Lorentz-invariant dispersion form for all dimension-≤ 4 operators [2]. Lorentz invariance therefore emerges in the long-wavelength limit, with violations suppressed by $(kL)^4$. Residual Lorentz violation enters only at fourth order in (kL), suppressed by $(E/E_P)^4 \sim 10^{-112}$ at optical frequencies — far below the experimental bound of 10^{-16} [6].

Resolution of the Collins et al. naturalness objection. Collins et al. [7] argued that any Lorentz-violating UV cutoff generates fine-tuning of order $\Lambda_{\rm UV}^2/M_P^2$ via radiative corrections. The SSM partially addresses this: the UV cutoff is the 2D boundary network, which carries *exact* continuous SO(2) symmetry. Radiative corrections in the boundary theory respect SO(2) at every loop order, and the exact RT map transmits this isotropic self-energy into the bulk without generating any Lorentz-violating dimension-4 operator. Heuristically, at the level of the boundary-bulk RT map, no counterterm appears to be generated. However, a complete argument requires an explicit calculation of the bulk effective action from the boundary theory — which has not been carried out — so this statement is tentative (see §10.4).

7 Schrödinger Equation as Cosserat Continuum Limit

7.1 The chiral Cosserat Lagrangian

Each lattice node carries a translational displacement u and a microrotation θ, the two independent fields of a three-dimensional Cosserat continuum [11] (three components each). The bond decomposition $K = S_{\rm trans} + S_{\rm tors} = 4 + 8$ refers to the two *bond subsets* that source these fields — the 4 tetrahedral bonds feeding the translational sector and the 8 remaining bonds feeding the rotational (torsional) sector — and is not a count of continuum degrees of freedom, which remain 3+3. We treat u and θ below as the coarse-grained scalar amplitudes of these two sectors. The Lagrangian density for a topological braid threading the lattice is:

$$\mathcal{L} = \tfrac{1}{2}(\partial_t u)^2 + \tfrac{1}{2}(\partial_t \theta)^2 - \tfrac{c^2}{2}(\nabla u)^2 - \tfrac{c^2}{2}(\nabla\theta)^2 - \tfrac{\omega_0^2}{2}(u^2+\theta^2) + \Omega(\dot{u}\theta - \dot{\theta}u). \tag{20}$$

The chiral coupling $\Omega(\dot{u}\theta - \dot{\theta}u)$ is *conjectured* to arise from the Berry connection of the discrete lattice wavefunction; a derivation of Ω from the $K = 12$ bond geometry has not been carried out (see §10.4).

7.2 Structural isomorphism with the Schrödinger equation

The Euler-Lagrange equations are:

$$\ddot{u} - c^2\nabla^2 u + \omega_0^2 u + 2\Omega\dot{\theta} = 0, \tag{21}$$

$$\ddot{\theta} - c^2\nabla^2\theta + \omega_0^2\theta - 2\Omega\dot{u} = 0. \tag{22}$$

Defining $\psi = u + i\theta$ and multiplying Eq. (22) by i and adding:

$$\frac{\partial^2\psi}{\partial t^2} + 2i\Omega\frac{\partial\psi}{\partial t} - c^2\nabla^2\psi + \omega_0^2\psi = 0. \tag{23}$$

Within this construction the complex unit i plays the role of the operator pairing translational and rotational modes of the Cosserat lattice: it is introduced as the bookkeeping combination $\psi = u + i\theta$ of two real fields, and acquires this geometric reading only once the chiral coupling Ω is present. We do not claim i emerges independently of that coupling, which is itself conjectural (§10.4). Substituting $\psi = \Phi\, e^{-i\Omega t}$ (Larmor frame) and taking the non-relativistic limit yields:

$$i\hbar\frac{\partial\Phi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\Phi. \tag{24}$$

On the role of $\hbar$. The Cosserat framework is a classical continuum theory: it does not intrinsically contain action quantization. The factor $\hbar$ enters as follows. Each FCC bond transmits at most one quantum per time step $\tau = 4L/c$ (from $c = 4v_{\text{lat}} = 4L/\tau$, Section 3). The natural energy quantum of the lattice is therefore $\varepsilon = \hbar/\tau = \hbar c/(4L)$, setting the Planck-scale bond energy scale. The Larmor substitution $\psi = \Phi\, e^{-i\Omega t}$ reduces the wave equation (23) to Schrödinger form when $\Omega = mc^2/\hbar$, i.e., when the rest energy mc^2 equals $\hbar\Omega \sim \hbar c/L$ — the bond energy scale. This identification is consistent with the soft threshold $m_{\text{soft}} = \hbar/(Lc)$ (Section 8), where the same scale reappears as the onset of lattice corrections. The content of this section is therefore a *structural isomorphism*: the long-wavelength continuum limit of the $K = 12$ Cosserat lattice takes the form of the free-particle Schrödinger equation once the chiral coupling is present. This is a non-trivial result — once the chiral coupling is present, its *form* (a translation–rotation cross term) is the generic Cosserat invariant and the value $\Omega = mc^2/\hbar$ and the Larmor substitution are then fixed by the reduction rather than inserted by hand — but it is an isomorphism, not an *ab initio* derivation of $\hbar$ from geometry alone, and the microscopic origin of Ω (its sign and magnitude from the $K = 12$ Berry connection) remains conjectural (§10.4).

8 The Two-Step Quantum-Classical Threshold

Two mass scales emerge from the $K = 12$ cuboctahedral unit cell. Both are fixed by $L = 1.843\,\ell_P$ (Section 5) and the exact geometry of the cuboctahedral faces, with no fitted parameters. We define them purely geometrically: $m_{\text{soft}} = \hbar/(Lc)$ is the mass at which the reduced Compton length $\lambda_c = \hbar/(mc)$ equals L, and $m_{\text{hard}} = \hbar\sqrt{3}/(Lc)$ is the mass at which λ_c equals the triangular face circumradius $L/\sqrt{3}$. The reduced Compton length

λ_c is the scale associated with the rest energy $mc^2 = \hbar c/\lambda_c$; it is *not* the spatial extent of the center-of-mass wavefunction in any experiment, which is set by trap frequencies and is many orders of magnitude larger.

Both thresholds are of order the Planck mass. Since $m_{\rm soft} = \hbar/(Lc) = m_P/(L/\ell_P) = m_P/1.843$ and $m_{\rm hard} = \sqrt{3}\, m_{\rm soft} = 0.940\, m_P$, with $m_P = \sqrt{\hbar c/G} \approx 21.8\,\mu$g, the scale itself is *not* a novel prediction: the Diósi-Penrose gravitational-collapse scale is likewise $\mathcal{O}(m_P)$. What is specific to the SSM, and what makes the prediction falsifiable, is the *structure* of the boundary — two thresholds rather than one, separated by the exact, parameter-free ratio $m_{\rm hard}/m_{\rm soft} = \sqrt{3}$ (the edge-to-circumradius ratio of the triangular face), with a concave-up profile in between — a quadratic onset $\Gamma \propto (m - m_{\rm soft})^2$ at $m_{\rm soft}$ that steepens to $\Gamma \propto [(m/m_{\rm soft})^2 - 1]^2$ approaching $m_{\rm hard}$. The remainder of this section develops this structure and contrasts it with the single smooth threshold of Diósi-Penrose.

Physical interpretation (conjectured). The SSM proposes that the lower geometric scale marks the onset of a reversible dispersive deformation of a center-of-mass superposition of mass m, and the upper scale the point at which coherent two-branch evolution can no longer be sustained. The reasoning is as follows: the Cosserat lattice dispersion (Eq. (26)) carries a correction $(kL)^2$. When this correction is evaluated at the wavenumber $k_c = mc/\hbar$ — the Compton scale, which is the natural scale at which a mass-m perturbation resolves the lattice in the gravitational sector — it becomes $(m/m_{\rm soft})^2$. The conjecture is that as this ratio crosses unity, the lattice can no longer maintain coherent two-branch evolution of the center-of-mass superposition. We emphasize that *this is a conjecture, not a derivation*: a first-principles calculation would require an explicit vacuum phonon bath Hamiltonian, a microscopic coupling to the center-of-mass degree of freedom, and a proper density of final states — none of which are constructed in this paper. We treat Eq. (30) as a phenomenological rate law motivated by the lattice geometry.

The central physical assumption. The weakest step in this chain is the choice of length scale, and we state it plainly. The thresholds are built from the *reduced Compton wavelength* $\lambda_c = \hbar/(mc)$, an internal rest-energy scale, whereas laboratory decoherence of a spatial superposition is governed by the *delocalization* Δx of the center-of-mass wavefunction, which is set by trap geometry and is orders of magnitude larger than λ_c for any realizable mass. The SSM posits that the lattice couples to the rest-energy scale of the superposed object, not to its delocalization — that the mass-m excitation "resolves" the lattice through λ_c in the gravitational sector, independent of how far apart the two branches sit. This is a physical assumption, not a theorem; it is the price of a parameter-free prediction, and it is what an experiment in the 11.8–20.5 μg window would test. We do not derive it from a microscopic system-bath model here, and the falsifiable content (the $\sqrt{3}$ ratio and the two-step profile) stands or falls with it.

8.1 The metric wall from cuboctahedral face geometry

The $K = 12$ unit cell is a cuboctahedron with 8 equilateral triangular faces of edge length L, established in Section 2. Every triangular face has a circumscribed circle of radius

$$r_{\rm min} = \frac{L}{\sqrt{3}}, \tag{25}$$

the distance from the face center to each of its three vertices. This is the circumradius formula for an equilateral triangle with side a: $R = a/\sqrt{3}$, applied with $a = L$. (Note:

the insphere radius of a regular tetrahedron with edge L is $L/(2\sqrt{6}) \approx 0.204L$, which is different; the relevant scale here is the circumradius of the triangular *face*, not a void.)

The physical meaning is direct. A probe at the centroid of a triangular face with Compton wavelength $\lambda_c = \hbar/(mc)$ shrinking to $r_{\min}$ simultaneously resolves all three face vertices at exactly bond-length separation. Further compression would require $\lambda_c < L/\sqrt{3}$: the probe would have to fit between nodes separated by L, which is the minimum bond length of the network. The lattice topology forbids it — this is the metric wall.

This length scale is not only analytic. The companion simulation [3] sweeps the exclusion radius R_{ex} of the growing network and finds a sharp geometric phase transition at $R_{\text{ex}} = L/\sqrt{3} \simeq 0.577\,L$, with a wide $K = 12$ stability plateau over $R_{\text{ex}} \in [0.58, 0.99]\,L$. Below $L/\sqrt{3}$ the exclusion is too weak and unphysical overlaps ($K_{\max} > 12$) appear; the transition therefore confirms, numerically, that $L/\sqrt{3}$ is the operative minimum-approach distance and not a tuned parameter. The two thresholds derived in Sections 8.2–8.3 inherit this scale directly.

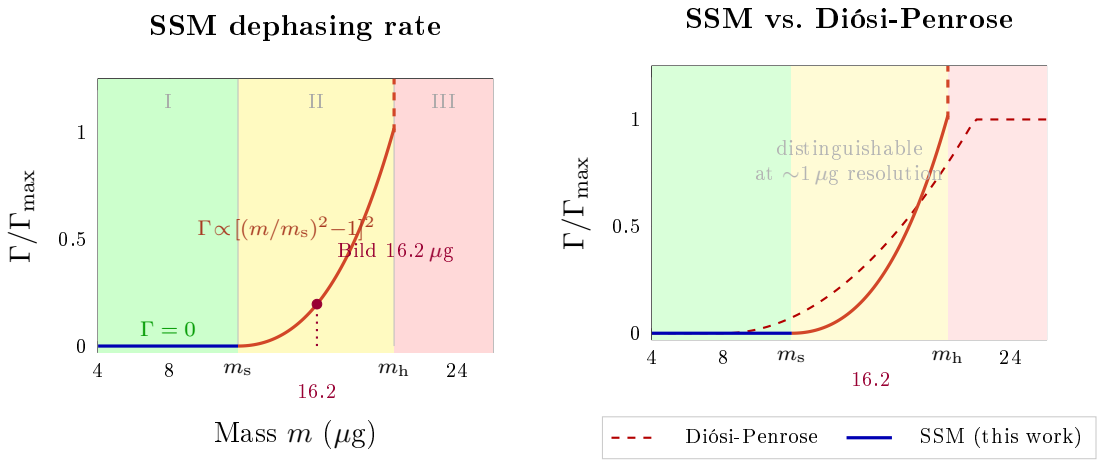


Figure 4: **Left:** SSM two-step dephasing rate $\Gamma(m)$ (normalized to $\Gamma_{\max}$). *Regime I* (green, $m < m_{\text{soft}} = 11.8\,\mu\text{g}$): $\Gamma = 0$, exact quantum mechanics. *Regime II* (yellow, $m_{\text{soft}} < m < m_{\text{hard}} = 20.5\,\mu\text{g}$): the rate rises as $\Gamma \propto [(m/m_{\text{soft}})^2 - 1]^2$ — a quadratic onset just above m_{soft} that steepens (concave up) toward m_{hard}. *Regime III* (pink, $m > m_{\text{hard}}$): topological cutoff, coherence impossible. Purple dot: Bild et al. $16.2\,\mu\text{g}$ resonator, at only ≈20% of the terminal rate (coherence survives). The regime-II curve is the *conditional* dephasing rate (present only if the lattice provides a phonon-bath coupling, Section 8.4); the bath-independent prediction in this window is the reversible dispersive deformation $\delta(m)$ of Eq. (31). **Right:** SSM (solid/thick) versus Diósi-Penrose (red dashed). The two models are distinguishable at $\sim 1\,\mu\text{g}$ resolution in the 10–25 μg window: DP predicts a single smooth threshold while SSM predicts a hard cutoff at $m_{\text{hard}} = \sqrt{3}\,m_{\text{soft}}$.

8.2 Soft threshold: onset of lattice corrections

Evaluating the Cosserat dispersion correction (Eq. (26)) at the Compton wavenumber $k_c = mc/\hbar$ (the scale at which rest energy equals $\hbar c/\lambda_c$):

$$E(k) = \frac{\hbar^2 k^2}{2m}\left[1 + \frac{(kL)^2}{6} + O\big((kL)^4\big)\right]. \tag{26}$$

For a center-of-mass state of mass m, the relevant wavevector is the Compton scale $k_c = mc/\hbar$, giving correction amplitude $(k_c L)^2 = (m/m_{\text{soft}})^2$, where the soft threshold is set by $(k_c L)^2 = 1$:

$$\boxed{m_{\text{soft}} = \frac{\hbar}{Lc} = \frac{\hbar}{1.843\,\ell_P\,c} \approx 11.8\ \mu\text{g}.} \tag{27}$$

Below m_{soft} the correction is $(m/m_{\text{soft}})^2 \ll 1$: the Schrödinger equation holds to all practical precision. Above it, the lattice is no longer transparent.

8.3 Hard threshold: circumradius cutoff

The metric wall at $r_{\min} = L/\sqrt{3}$ gives the hard mass scale: the Compton wavelength that fits exactly inside the circumscribed circle of a triangular face:

$$\lambda_c = \frac{L}{\sqrt{3}} \implies \boxed{m_{\text{hard}} = \frac{\sqrt{3}\,\hbar}{Lc} = \sqrt{3}\,m_{\text{soft}} \approx 20.5\ \mu\text{g}.} \tag{28}$$

The ratio $m_{\text{hard}}/m_{\text{soft}} = \sqrt{3}$ is exact. It comes from the edge-to-circumradius ratio of an equilateral triangle ($L : L/\sqrt{3} = \sqrt{3}$) and is independent of $\hbar$, c, or G. At m_{hard}, the wavepacket spans the circumscribed circle of the face exactly; any further compression would require $\lambda_c < L/\sqrt{3}$, placing it within the triangular face while all three vertices are closer than the bond length. The SSM conjectures that no lattice configuration can sustain the superposition past this point; this is the hard coherence cutoff. Within the window the rate approaches its terminal value $4\,\Gamma_0$ continuously (Eq. (30)); the cutoff marks not a divergence of that rate but the point at which the perturbative expansion in (kL) breaks down ($\delta = \frac{1}{2}$) and the geometry can no longer contain the superposition. A caveat on the domain of validity is essential here, and we state it plainly. The small-(kL) expansion of Eq. (26) is, by construction, controlled only for $(k_c L)^2 \ll 1$, i.e. for $m \ll m_{\text{soft}}$. The window $m_{\text{soft}} < m < m_{\text{hard}}$ corresponds to $1 < (k_c L)^2 < 3$, where the expansion is no longer a convergent approximation but an extrapolation. We use it in this range only to locate the geometric endpoints $(k_c L)^2 = 1$ and $(k_c L)^2 = 3$ — the bond length and the face circumradius — and not as a quantitatively reliable profile. The $\sqrt{3}$ ratio between the endpoints is fixed by exact triangle geometry and does not depend on the expansion; the shape of the rate between them does, and is therefore heuristic. A microscopic operator-level derivation of this breakdown is deferred to future work.

8.4 Dephasing rate from dispersion correction

Treating the dispersion correction as a coupling to a lattice phonon bath (whose modes, coupling Hamiltonian, and density of states are not derived microscopically here), and modeling the resulting dephasing phenomenologically via Fermi's golden rule:

$$\Gamma = \frac{2\pi}{\hbar}\,|\langle f|\,\delta\hat{H}\,|i\rangle|^2\,\rho(E_f), \tag{29}$$

Here $\delta\hat{H}$ denotes the lattice correction to the kinetic operator and $\rho(E_f)$ the density of final bath states. The correction amplitude is $\delta H/H_0 = (m/m_{\text{soft}})^2/6$. We model the dephasing matrix element as the *excess* of this amplitude over its soft-threshold value, $|\langle f|\delta\hat{H}|i\rangle|^2 \propto [(m/m_{\text{soft}})^2 - 1]^2$ across the window. The subtraction of unity is an imposed modeling choice, not a consequence of the amplitude above: it enforces $\Gamma = 0$ at m_{soft} so

that the rate switches on at the geometric threshold rather than below it. With $\rho(E_f) \propto m$ slowly varying:

$$\Gamma_{\text{lattice}}(m) \;=\; \begin{cases} 0 & m \le m_{\text{soft}}, \\ \Gamma_0 \left[\left(\dfrac{m}{m_{\text{soft}}} \right)^2 - 1 \right]^2 & m_{\text{soft}} < m < m_{\text{hard}}, \\ \text{coherence not sustained} & m \ge m_{\text{hard}}, \end{cases} \tag{30}$$

where $\Gamma_0 = \hbar/(mL^2)$. The profile is not linear and does not taper off near the upper threshold; it is *concave up*. Just above m_{soft} it reduces to the quadratic onset $\Gamma \approx 4\Gamma_0\,\varepsilon^2$, $\varepsilon = (m - m_{\text{soft}})/m_{\text{soft}}$, but it then steepens to a quartic dependence on mass and rises to its terminal value $\Gamma_0[(m_{\text{hard}}/m_{\text{soft}})^2 - 1]^2 = 4\,\Gamma_0$ at m_{hard}, where the geometric cutoff (Section 8.1) ends the window. Equivalently, the deterministic deformation $\delta(m)$ (Eq. (31)) grows quadratically in mass ($\frac{1}{6} \to \frac{1}{2}$) and the rate grows as the square of its excess over the soft-threshold value. Concretely: at the window midpoint in mass the rate has reached only $\sim$20% of its terminal value (the Bild point, $m/m_{\text{soft}} \approx 1.37$, sits at $\approx 20\%$), then climbs to 39%, 61%, and 89% at $m/m_{\text{soft}} = 1.5$, 1.6, 1.7 before the cutoff. Coherence therefore stays near-perfect through most of the window and collapses increasingly rapidly as $m \to m_{\text{hard}}$ — the opposite of a uniform linear decay. The *normalized* profile (the percentages above) is fixed by the matrix element $[(m/m_{\text{soft}})^2 - 1]^2$ and is therefore independent of the bath details; only the absolute scale Γ_0 depends on the (here undetermined) lattice coupling, so the falsifiable predictions are the shape and the two threshold masses, not an absolute coherence time.

What happens between the thresholds. The leading lattice term in Eq. (26) is Hermitian, so its first effect is *unitary*: it modifies the dispersion of the center-of-mass mode and therefore deforms the wavepacket deterministically rather than collapsing it. The size of the deformation is the dimensionless correction to the kinetic term at the Compton scale,

$$\delta(m) \;=\; \frac{1}{6}\left(\frac{m}{m_{\text{soft}}} \right)^2, \tag{31}$$

which grows from $\delta = 1/6 \approx 17\%$ at $m = m_{\text{soft}}$, through $\delta \approx 31\%$ at the Bild mass 16.2 μg, to exactly $\delta = 1/2$ at $m = m_{\text{hard}} = \sqrt{3}\,m_{\text{soft}}$. At m_{hard} the lattice correction reaches half the bare kinetic term — the Compton wavelength has shrunk to the metric wall $L/\sqrt{3}$, the perturbative expansion in (kL) ceases to hold, and the geometry can no longer contain the superposition. Physically, the SSM therefore predicts a sequence, not a single event (Fig. 4): below m_{soft} the wavepacket evolves under exact quantum mechanics; between the thresholds it suffers a mass-dependent, deterministic, and in-principle reversible dispersive deformation of size $\delta(m)$; only at m_{hard} does coherence become geometrically impossible. Whether the deformation also seeds genuine (irreversible) decoherence depends on coupling to lattice modes — the dephasing rate Eq. (30) describes that channel *if* the coupling exists, but the deterministic deformation $\delta(m)$ is the robust, bath-independent prediction. The deformation is quoted at the Compton scale; translating it into an observable distortion of the interference pattern requires the Compton-coupling assumption stated above.

Consistency with Bild et al. The 16.2 μg resonator [9] sits between the thresholds. The SSM predicts a deterministic dispersive deformation of size $\delta = \frac{1}{6}(16.2/11.8)^2 \approx 31\%$ at the Compton scale (Eq. (31)), but the topological cutoff at m_{hard} has not been reached, so the superposition remains coherent — consistent with the observation of cat states at

this mass. This is not a post-hoc fit: m_{soft} and m_{hard} follow from L, fixed independently in Section 5.

Distinction from Diósi-Penrose. The difference is qualitative, not just a matter of one threshold versus two. In the DP model [8] the gravitational self-energy E_Δ of the mass difference between the two branches drives a *stochastic, irreversible* collapse at rate $\tau^{-1} = E_\Delta/\hbar$; the effect depends on the branch separation Δx and the mass distribution, sets in smoothly for all masses, and carries the free length parameter R_0. The SSM predicts instead a *deterministic* dispersive deformation $\delta(m)$ (Eq. (31)) that depends only on the object's mass m through λ_c — not on Δx — is exactly zero below m_{soft}, and terminates at a hard geometric cutoff at $\sqrt{3}\,m_{\text{soft}}$, with no fitted parameters. A single experiment discriminates the two: hold Δx fixed and scan the mass. DP predicts a smooth, Δx-dependent loss of fringe visibility with no special mass; the SSM predicts a flat null below $11.8\,\mu$g, a deterministic onset $\propto (m - m_{\text{soft}})^2$ that is independent of Δx, and a sharp cutoff at $20.5\,\mu$g. The two are distinguishable with $\sim 1\,\mu$g mass resolution in the 10–25 μg window.

9 Comprehensive Experimental Comparison

Table 1 collects the foundational predictions of this paper against the best available measurements. Each quantity follows from the $K = 12$ FCC lattice via the chain in Section 10.2, together with the additional assumptions that chain makes explicit; none involves a fitted parameter.

Observable	SSM derivation	SSM value	Measured
Foundational (this paper)			
c/v_{lat}	$S_{\mu\nu} = 4\delta_{\mu\nu}$, $K = 12$	4 (exact)	4
G_N	imposed as ℓ_P^2 (RT input)	ℓ_P^2	ℓ_P^2
L	RT + $G_N = \ell_P^2$	$1.843\,\ell_P$	—
Lorentz violation	$(E/E_P)^4$ suppressed	$\sim(E/E_P)^4$	$< 10^{-16}$ [6]
m_{soft}	$\hbar/(Lc)$, $L = 1.843\,\ell_P$	$11.8\,\mu$g	consistent [9]
m_{hard}	$\sqrt{3}\,m_{\text{soft}}$	$20.5\,\mu$g	$> 16.2\,\mu$g tested [9]
$m_{\text{hard}}/m_{\text{soft}}$	circumradius/edge of $\triangle$	$\sqrt{3}$ (exact)	untested
Schrödinger eq.	Cosserat structural isomorphism	isomorphic	postulated

Table 1: Foundational SSM predictions versus experiment. The quantities involve no fitted parameters, but each rests on structural assumptions beyond $K = 12$ alone (the RT relation, a single propagation speed, the chiral coupling, and the Compton-coupling hypothesis of Section 8), collected in Section 10.4. A dash under "Measured" indicates a Planck-scale quantity not directly accessible.

Three entries warrant comment. The ratio $m_{\text{hard}}/m_{\text{soft}} = \sqrt{3}$ is the sharpest near-term

prediction: it requires only two experiments bracketing the 11.8–20.5 μg window, calibrated against each other, with no reference to any Standard Model parameter. Current acoustic-resonator technology is within a factor of two of this range [9]; MAQRO [10] could reach it within the decade. The Lorentz-violation prediction is consistent with current limits. The model suppresses Lorentz-violating bulk operators by $(E/M_P)^4$, which is of order 10^{-112} at optical frequencies. This is a raw suppression factor rather than a specific Standard-Model-Extension coefficient, so it is not directly the same quantity as the tabulated bound of order 10^{-16}; the point is only that the predicted effect lies roughly 90 orders of magnitude below any current sensitivity. The Schrödinger entry is unusual: the "measured" column records what quantum mechanics assumes as a postulate, while the SSM column records a structural isomorphism — the $K = 12$ Cosserat continuum limit takes the form of the free-particle Schrödinger equation once a chiral translation-rotation coupling is supplied, with the imaginary unit i entering through that coupling rather than being posited separately. The coupling itself is conjectured, not derived (§10.4).

10 Discussion

10.1 Why FCC and not another lattice?

The FCC lattice is selected by three requirements: (a) it saturates the Kepler bound ($K = 12$) [4]; (b) its four {111} families generate emergent SO(3), which no other 3D Bravais lattice achieves; and (c) unlike the simple-cubic and BCC lattices, it is non-bipartite, which modifies where fermion doublers appear in the Brillouin zone compared to standard Wilson-fermion constructions. Whether the SSM Dirac operator (6) fully evades the Nielsen-Ninomiya theorem [13] — which applies to all translation-invariant local Dirac operators, regardless of whether the lattice is bipartite — has not been demonstrated and is listed as an open problem in §10.4.

10.2 Derivation chain summary

The logical dependencies are:

1. **Input:** $K = 12$ FCC lattice with Bell-pair bonds.
2. $c/v_{\text{lat}} = 4$ (Section 3): the ratio of macroscopic wave speed to lattice update rate equals the structure tensor eigenvalue $S_{\mu\nu} = 4\delta_{\mu\nu}$, exact from $K = 12$.
3. $L = 1.843\,\ell_P$ given $G_N = \ell_P^2$ (Sections 4–5): with $G_N = \ell_P^2$ imposed, the RT relation fixes L uniquely; the energy-density bound ($L > 0.77\,\ell_P$) and an energy-balance minimum ($L_\star \sim \ell_P$, Section 4.2) independently confirm the Planck *scale* (not the coefficient 1.843), which RT supplies. The application of RT to this lattice is itself an assumption.
4. Emergent SO(3,1) (Section 6): isotropic dispersion below $O((kL)^4)$, established in [2]; reviewed here. A single propagation speed for all modes is assumed, not derived.
5. Schrödinger equation (Section 7): from the chiral Cosserat Lagrangian, conditional on the conjectured coupling Ω.
6. $m_{\text{soft}} = 11.8\,\mu$g, $m_{\text{hard}} = 20.5\,\mu$g (Section 8): from L and the metric wall $r_{\min} = L/\sqrt{3}$, under the Compton-coupling hypothesis.

10.3 Comparison with other discrete spacetime approaches

Feature	SSM	LQG	Causal Sets	Lattice QCD
c/v_{lat} ratio?	Yes ($= 4$, exact)	No	No	No
G_N derived?	Consistency cond. (RT)	No	No	No
ℓ_P fixed?	Yes (via RT)	Assumed	No	$a \to 0$
Emergent Lorentz?	$(E/E_P)^4$ at ℓ_P	Debated	Statistical	$(pa)^2$ at tunable a
Decoherence?	Two-step, $\sqrt{3}$	Various	Not derived	N/A
Fitted parameters	0	Several	1 (ρ)	Several

10.4 What remains open

The following foundational questions are left to future work and represent limitations of the present paper.

Vacuum Hamiltonian. The Bell-pair bond structure is a static structural input. No Hamiltonian is given from which $|\Phi^+\rangle$ on every bond emerges as a ground state; stabilizing this vacuum dynamically is an open problem.

Chiral coupling Ω in the Cosserat Lagrangian. The term $\Omega(\dot{u}\theta - \dot{\theta}u)$ in Eq. (20) is asserted to arise from the Berry connection of the lattice wavefunction but is not derived. A derivation of the precise value of Ω from the $K = 12$ bond geometry has not been carried out.

Phase-transition dynamics. The $K{=}6 \to K{=}4 \to K{=}12$ sequence in §2 is now supported numerically: the companion simulation [3] exhibits the transition kinematically and confirms $K = 12$ bulk saturation under finite-size scaling. What remains open is a closed-form free energy — an action or partition function from which the transition follows analytically — rather than the existence of the transition itself.

Decoherence mechanism. The rate law (Eq. (30)) and the hard cutoff at m_{hard} are phenomenological conjectures. A microscopic derivation requires an explicit bath Hamiltonian, conjugate coupling operator, density of final states, and a demonstration that coherence-protecting symmetry is broken at the topological cutoff.

Nielsen-Ninomiya evasion. The Nielsen-Ninomiya theorem [13] states that any local, translation-invariant, Hermitian lattice Dirac operator has equal numbers of left- and right-handed modes, regardless of whether the lattice is bipartite. The SSM is non-bipartite, which modifies the locations of fermion doublers in the Brillouin zone, but a proof that the tensor-network Dirac operator derived in Section 3 satisfies the conditions of the theorem — or that those conditions are not met — has not been given.

Single propagation speed. The step from emergent SO(3) to emergent SO(3,1) assumes all low-energy Cosserat modes share a single speed c. A proof that the $K = 12$ lattice enforces this (rather than admitting different group velocities for different modes) has not been given.

Two-step threshold prediction. The ratio $m_{\text{hard}}/m_{\text{soft}} = \sqrt{3}$ is the cleanest falsifiable prediction; it depends on no Standard Model input. The Standard Model gauge group, quark masses, and cosmological parameters are addressed in separate papers.

10.5 Falsifiability

- **Planck-suppressed Lorentz violation:** the model predicts violations at order $(E/E_P)^4 \sim 10^{-112}$ at optical frequencies, far below current bounds. Any detection of birefringence or time-of-flight anomalies at energies well below E_P would falsify the model.
- **Two-step structure** separated by exactly $\sqrt{3}$, with an exactly null response below $m_{\text{soft}} = 11.8\,\mu\text{g}$ and a hard cutoff at $m_{\text{hard}} = 20.5\,\mu\text{g}$ — not the single smooth threshold of Diósi-Penrose.
- **Deterministic deformation** between the thresholds: a reversible dispersive distortion $\delta(m) = \frac{1}{6}(m/m_{\text{soft}})^2$ that depends on the mass alone and not on the branch separation Δx; a mass scan at fixed Δx separates this from the Δx-dependent stochastic collapse of Diósi-Penrose.
- G_N **from entanglement:** precision short-distance measurements of Newton's constant provide an independent test.

11 Conclusion

From the single structural choice $K = 12$ FCC with Bell-pair bonds, together with the geometric and holographic assumptions made explicit in the text, four results follow with no fitted parameters:

1. $c/v_{\text{lattice}} = 4$ from the cuboctahedral structure tensor $S_{\mu\nu} = 4\delta_{\mu\nu}$: the macroscopic speed of light is exactly four times the lattice update rate.
2. $L = 1.843\,\ell_P$ from the Ryu-Takayanagi relation $G_N = \sqrt{2/3}\,L^2/(4\ln 2)$ on the hexagonal boundary, once $G_N = \ell_P^2$ is imposed. This is a consistency condition fixing the bond length, not a first-principles derivation of G_N.
3. A two-step quantum-to-classical boundary: a reversible dispersive deformation onsets at $m_{\text{soft}} = 11.8\,\mu\text{g}$ and coherence becomes geometrically unsustainable at $m_{\text{hard}} = 20.5\,\mu\text{g}$, with ratio exactly $\sqrt{3}$ from the circumradius-to-edge ratio of the cuboctahedral triangular face.
4. A structural isomorphism between the long-wavelength $K = 12$ Cosserat continuum and the Schrödinger equation, conditional on a chiral translation-rotation coupling of the assumed form: the imaginary unit i then enters through that coupling rather than as a separate postulate. The coupling is conjectured, not derived.

Emergent Lorentz invariance (Planck-suppressed violations at order $(E/E_P)^4$), established in the published framework [2], is reviewed and the Collins et al. naturalness objection is partially addressed; a complete proof requires explicit bulk effective-action calculation from the boundary theory.

The comparison in Table 1 shows the framework is consistent with all available experimental data; its central prediction — the two-step threshold with ratio $\sqrt{3}$ — remains untested and is the natural target for the next generation of macroscopic-superposition experiments.

Author Contributions

Raghu Kulkarni: Conceptualization; Methodology; Formal analysis; Investigation; Writing — original draft; Writing — review & editing; Visualization; Funding acquisition.

Data Availability

No datasets were generated or analyzed during this study. All results follow analytically from the geometric framework described in the text.

Declaration of Competing Interests

The author declares no competing interests.

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